Name_____ Date_____ Period_____

LESSON 2-9 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Finding Mean Absolute Deviation (MAD)

Maryland Standard 6.SP.B.5c · Learning target – I can find the mean absolute deviation of a data set and explain what it measures.

REMEMBER

- The **MAD** (mean absolute deviation) is the average **distance** of the values from the mean. Distances are never negative.
- Find the **mean**. Find each value's **distance** from the mean. Then find the **mean of the distances**.

WORKED EXAMPLE

Ruth's quiz scores are 6, 8, 9, and 13. Find the MAD.

Step 1 Mean: 36 ÷ 4 = 9

Step 2 Distances from 9: 3, 1, 0, 4

Step 3 Mean of distances: 8 ÷ 4 = 2

Answer **MAD = 2 points**

WORD BANK

mean – add, then divide by how many

distance – how far apart; never negative

MAD – the mean of the distances from the mean

PRACTICE

show your work · label units

1. The mean of 5, 7, 10, 12, and 16 is 10. Complete the table. Then find the MAD.

Step 1 Find how far each value is from 10.

Step 2 Add the distances.

Step 3 Divide by 5, the number of values.

Value	**5**	**7**	**10**	**12**	**16**
Distance from 10					

MAD_____

2. The mean is given. Find the MAD.

Step 1 Find each distance from the mean.

Step 2 Add the distances. Divide by how many values.

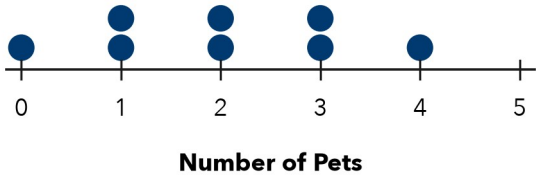
a. 1, 3, 5, 7 (mean 4) _____

b. 20, 25, 35, 40 (mean 30) _____

c. 6, 6, 6, 6, 6 (mean 6) _____

d. 2, 5, 8, 9, 11 (mean 7) _____

3. Pets. The dot plot shows the number of pets 8 students have.



a. Mean: _____ pets

b. MAD: _____ pets

4. Wait Times. Six patients **waited** 10, 14, 15, 17, 18 and 22 minutes. Find the **mean**. Then find the **MAD**.

Show your work.

Mean and MAD_____

5. Find the Error. For 3, 5, 9, 11, 12 (mean 8), Eli found the distances 5, 3, 1, 3, 4. Their sum is 16. He wrote MAD = 16 ÷ 8 = 2. Explain his error and find the MAD.

Frame *Eli divided by ___ but he should divide by ___. The MAD is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home what a MAD of 0 would mean for a set of test scores.

Name_____ Date_____ Period_____

LESSON 2-9 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Comparing Data Sets Using MAD

Maryland Standard 6.SP.B.5c · Learning target – I can compare two data sets that share a mean by comparing their mean absolute deviations.

REMEMBER

- When two data sets have the **same mean**, compare their MADs. State both numbers.
- A **smaller MAD** means values are closer to the mean: more **consistent**. A larger MAD means more **variability**.

WORKED EXAMPLE

Baker A's loaves: 496, 500, 500, 504 g. Baker B's: 488, 496, 504, 512 g. Both means are 500 g. Who is more consistent?

Step 1 A's distances: 4, 0, 0, 4. MAD = 8 ÷ 4 = 2 g

Step 2 B's distances: 12, 4, 4, 12. MAD = 32 ÷ 4 = 8 g

Answer **Baker A (MAD 2 g versus 8 g)**

WORD BANK

consistent – values stay close together

MAD – typical distance from the mean

variability – how spread out the values are

PRACTICE

show your work · label units

1. **Basketball.** Both players have a mean of 12 points. **Kai:** 10, 11, 12, 13, 14. **Leo:** 4, 8, 12, 16, 20.

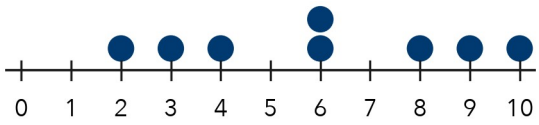
Step 1 Find each distance from 12 for Kai. Average them.

Step 2 Do the same for Leo.

Step 3 The smaller MAD is more consistent.

a. Kai's MAD: ____ **b.** Leo's MAD: ____ **c.** More consistent: ____

2. **Summer Reading.** The dot plot shows the books read by 8 students in Club B. Club A has a mean of 6 books and a MAD of 0.5 book.



Books Read (Club B)

a. Club B mean: _____

b. Club B MAD: _____

c. More spread out: _____

3. **Tomatoes Picked.** Both means are 20. **Garden A:** 18, 19, 21, 22. **Garden B:** 12, 16, 24, 28.

Step 1 Find each distance from 20.

Step 2 Add and divide by 4 for each garden.

a. A's MAD: ____ **b.** B's MAD: ____ **c.** More consistent: ____

4. **Bus Routes.** Both routes take **24 minutes** on average. Route 5 has **MAD 1.5**. Route 9 has **MAD 6**. Which route is **more predictable**?

☐ Route 5 ☐ Route 9 ☐ They are the same

5. **Find the Error.** Jordan says Set R (49, 50, 50, 51) must have a greater MAD than Set S (2, 6, 14, 18) because its numbers are bigger. Find both MADs and explain.

Frame *Jordan is wrong because bigger numbers do not mean ___. Set S has the larger MAD: ___.*

TEST PRACTICE

choose the best answer

6. Sets A and B both have a mean of 40. A's MAD is 2.5. B's MAD is 8. Which is true?

A A varies less. **B** B varies less. **C** Same spread. **D** Cannot tell.

7. What is the MAD of 3, 5, 7, 9?

A 0 **B** 2 **C** 6 **D** 8

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___